

DATA200

Final Presentation

Group 6: Screentime Bots

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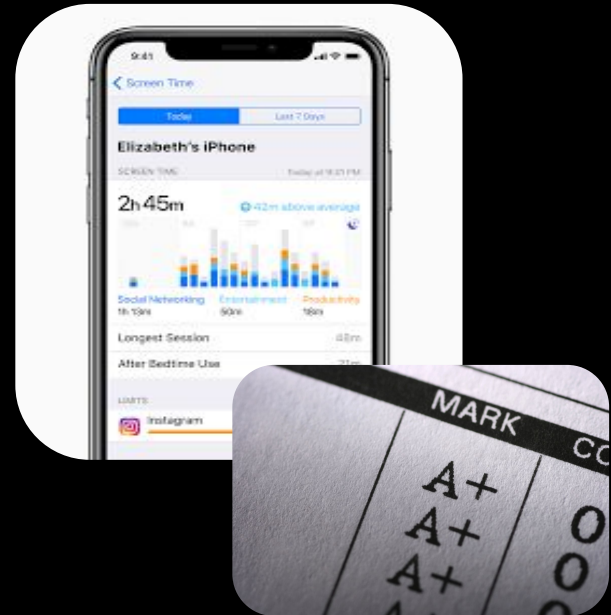
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Project Topic: Impact of Screen Time on Academic Performance

We used survey data from students to analyze the impact of their screen time on academic performance.

We hoped to determine whether students who spend less time on their screen maintain high GPA levels, suggesting a correlation between screen time and academic outcomes.

We also assessed what kinds of applications their screen time consists of, as well as their intentionality as they use their phones. This allowed us to consider factors of phone usage that impact academic performance.



Why We Chose This Topic

RELEVANCE

Many college students have high screen-time on their devices as a result of academic necessity, social communication, and entertainment. However, for many, it's become habitual use.

SIGNIFICANT INSIGHTS

Understanding the relationship between screen-time and GPA can help students ascertain and improve their academic performance.

SIMPLE DATA COLLECTION

College students' GPA, major, and their screen-time are relatively easy variables for us to collect.



Ethical Topics We Focused On

DATA OWNERSHIP

We obtained informed consent from survey respondents, in order to own their data for the purpose of academic research.

DATA ANONYMITY

Participants were not asked to provide personal identifying information. Hence, data is not be linked to specific individuals.

DATA PRIVACY

Participants know how their data will be collected, used, and shared. It will strictly remain within this research project.

DATA VALIDITY

Typed responses were reviewed to correct spelling errors and ensure information was entered in the correct format.

Data Collection

1. CREATE SURVEY

- Ask for informed consent.
- Questions and response formats are constructed to collect only valid data types and ranges.
- Questions are formatted in a way to avoid bias
- Question order prioritizes data numerical data collection before opinion based data.

2. DETERMINE AUDIENCE

- Determined that an audience consisting of students from varying years and majors at UMD would be representative the population of UMD students.

3. SEND OUT SURVEY

- Reach out to a diverse selection of students with varying majors and graduation dates.
- Distribution primarily through social media and individual outreach.

4. DATA VALIDATION

Collected 67 responses from over a month to ensure a diverse sample.

Used required fields to avoid incomplete responses. Minimized response bias by keeping the survey short.

Survey

We conducted a survey to assess the influence of screen time on academic performance, and shared it with fellow college students directly through messages and on social media.

We collected 67 responses from this survey in total.

https://docs.google.com/forms/d/1kAqno_gUBBW9BTXY0Ki8gXz11xs10R0-ldr4PcjiZU8A/edit



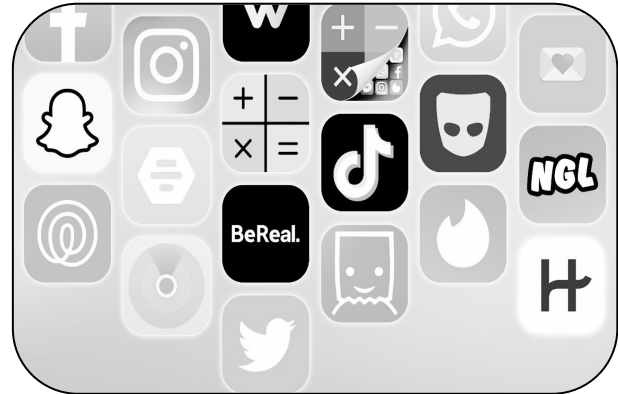
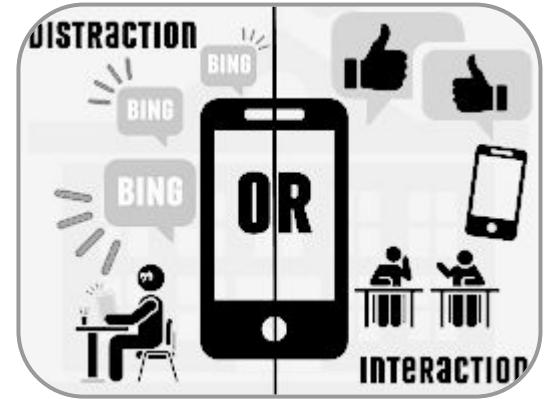
Impact of Screen Time on Academic Performance

Your participation in this survey is completely voluntary and your data will remain confidential. You will not be asked to provide personal identifying information so you will remain anonymous. Your information will be used to analyze the impact of screen time on academic performance. The information you provide will be used solely for academic purposes as part of our DATA 200 course project, which focuses on analyzing the impact of screen time on academic performance.

Survey Changes

Types of Questions Added

- Frequency of phone usage
- Time limits on apps
- Ranking most used app categories (eg. social media, education, protectivity, health, etc.)
- Listing most used apps
- Academic year and graduation year
- STEM non-STEM major
- Phone distraction during study time



Data Processing

Data Cleaning

- Reviewed all survey responses to ensure consistency

Reasoning

- Ensured GPA data was in the correct numeric format to allow accurate comparison

Other Measures

- Check and Removed missing responses or incomplete entries

Analysis

Research Focus	Method	Visualization
GPA vs. Screen Time	Spearman correlation	Boxplot / Bar chart
GPA vs. Meaningfulness	Regression	Scatterplot
Study Time vs. Screen Time	Spearman	Stacked bar chart
Credits vs. Screen Time	Chi-squared	Grouped bar chart
Addiction vs. Study Time	Spearman	Bar chart
Addiction vs. Purposefulness	Spearman	Heatmap or scatterplot

1. Does higher screen time correlate with lower GPA?

Variables:

- Q1: Average daily screen time (ordinal)
- Q12: Semester GPA (numeric)

Method:

- Spearman correlation (since screen time is ordinal)

Visualization:

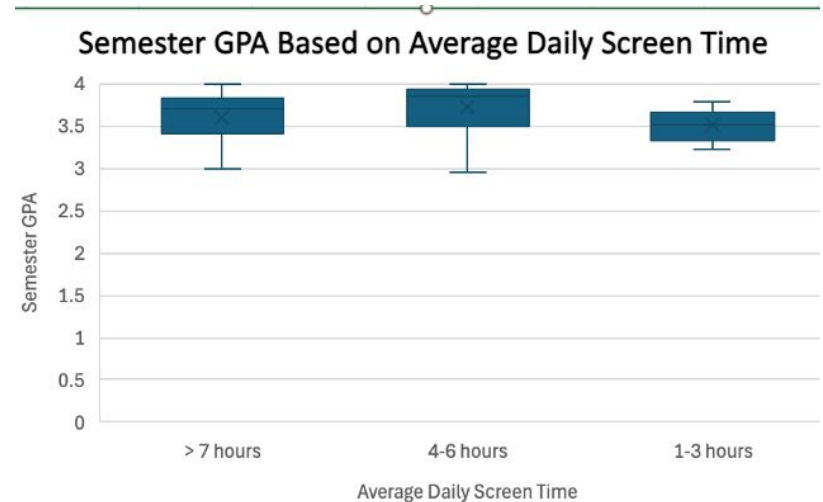
- Grouped box plots: Average GPA per screen time category

Spearman Correlation

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Spearman Correlation: -0.019  
P-value: 0.878
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- A Spearman correlation was performed to assess the relationship between average daily screen time and GPA
- P-value > 0.05 $\rightarrow$ not statistically significant

Interpretation: Average Screen Time was not significantly correlated with Semester GPA in this dataset



2. Does meaningful screen time correlate with GPA?

Variables:

- Q4: Screen time purposefulness (Likert scale)
- Q12: Semester GPA

Method:

- Spearman correlation or linear regression
- Check if more meaningful usage is associated with higher GPA

Visualization:

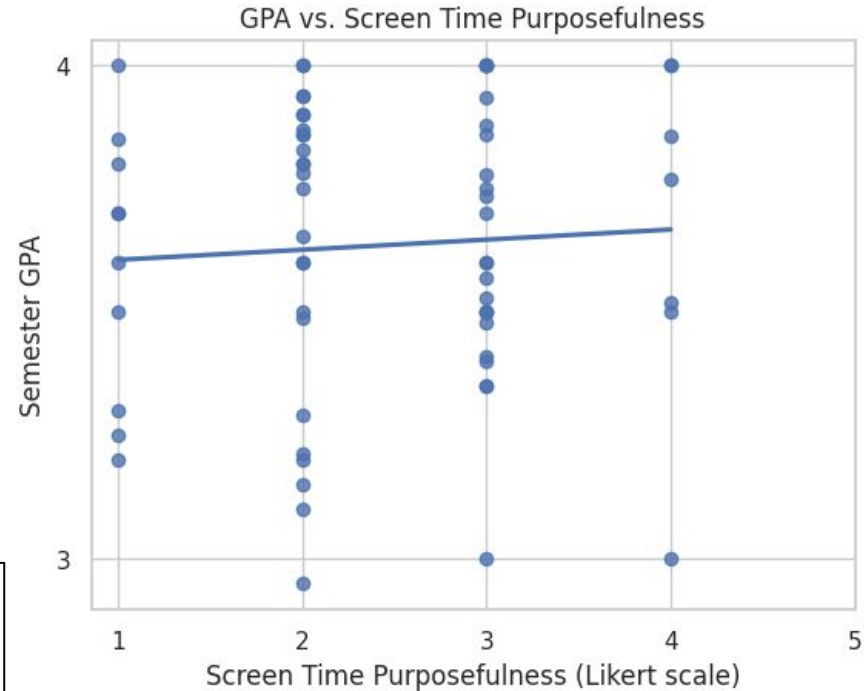
- Scatterplot with regression line (GPA vs. Screen time purposefulness)

Spearman Correlation

Spearman Correlation Coefficient: 0.06
P-value: 0.6293

- A Spearman correlation was performed to assess the relationship between GPA and student's reported score of screen time purposefulness.
- P-value > 0.05 → not statistically significant

Interpretation: No significant correlation according to the calculations. However, the scatterplot shows a slight upward trend, which may hint at a weak positive association—though this pattern is not strong enough to be conclusive.



3. Do students who study more have lower screen time?

Variables:

- Q14: Average weekly study hours (ordinal)
- Q1: Average daily screen time (ordinal)

Method:

- Spearman correlation

Visualization:

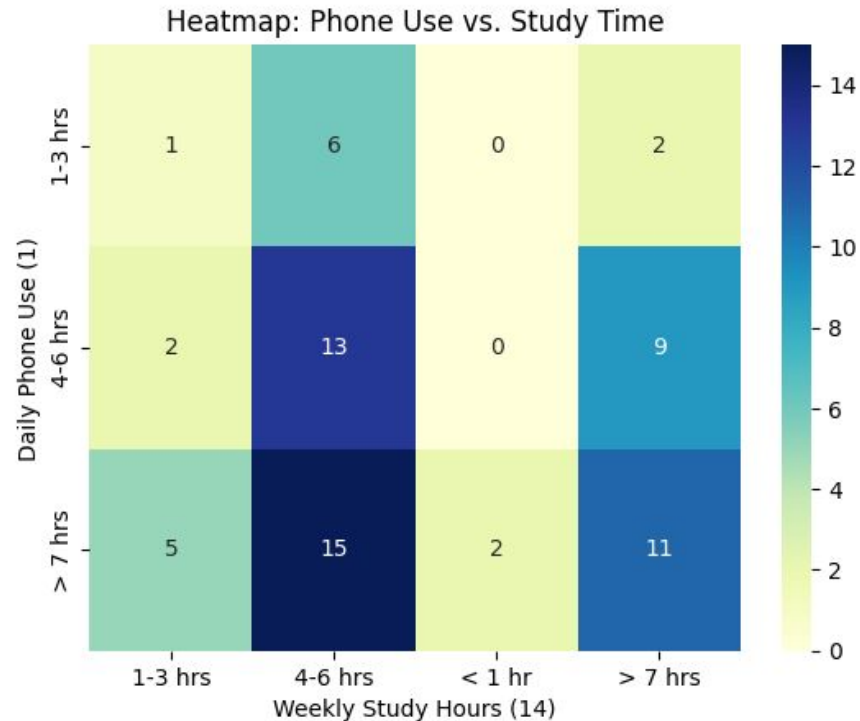
- Heatmap for phone use vs study time

Spearman Correlation

Spearman correlation coefficient: -0.0519
P-value: 0.6787

- A Spearman correlation was performed to assess the relationship between study time and average daily screen time
- P-value > 0.05 → not statistically significant

Interpretation: Study time was not significantly correlated with Average Screen Time in this dataset



4. Do students with more credits take have lower screen time?

Variables:

- Q11: Credits taken last semester (ordinal)
- Q1: Average daily screen time (ordinal)

Method:

- Chi-squared test

Visualization:

- Grouped box plots: Screen time category vs. credits group

Chi-Squared test

Expected Frequencies Table:

Screen Time	1-3 hours	4-6 hours	>7 hours
Credits			
11-13	0.830769	2.123077	3.046154
12-15	4.569231	11.676923	16.753846
<11	0.138462	0.353846	0.507692
>15	3.461538	8.846154	12.692308

- A Chi-Square test was performed to assess the relationship between number of credits taken and average daily screen time
- Students taking 12-15 credits reported the highest levels of average screen time

Interpretation: Although patterns are visible, there is not statistical evidence to support a relationship between credits and screen time.

Explanation for Result: Since most respondents took 12-15 credits, trends in screen time are more influenced by this group

5. How does weekly study time relate to phone addiction?

Variables:

- Q5: Phone addiction (Likert scale)
- Q14: Average weekly study hours (ordinal)

Method:

- Spearman correlation

Visualization:

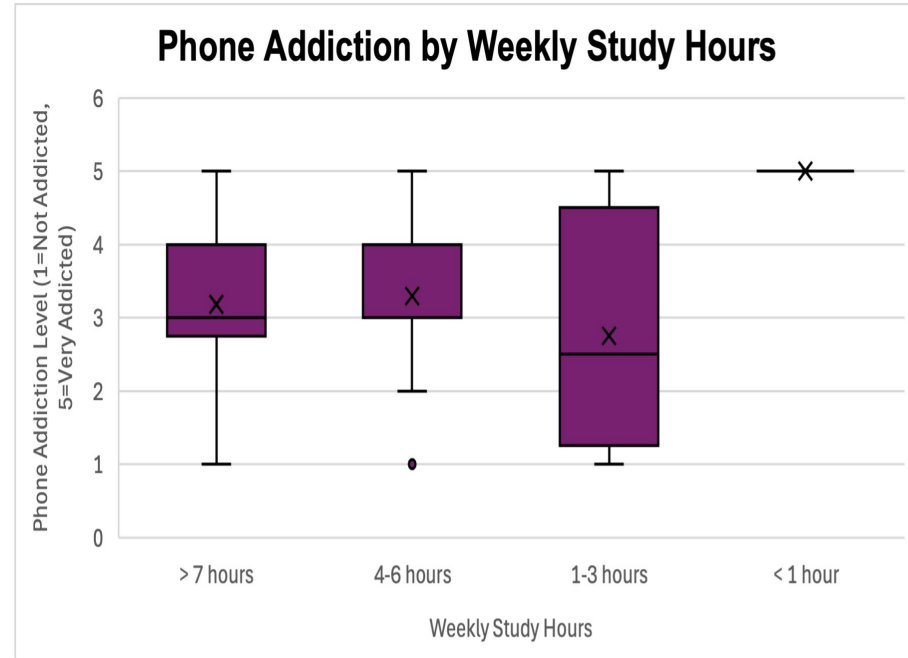
- Grouped box plots: Phone addiction by Weekly Study Time

Spearman Correlation

↔ Spearman Correlation (ρ): 0.354
P-value: 0.559

- A Spearman correlation was performed to assess the relationship between weekly study hours and phone addiction levels
- P-value $> 0.05 \rightarrow$ not statistically significant

Interpretation: An insignificant relationship was found between study time and phone addiction, however, no strong evidence supports a meaningful relationship in this dataset.



6. Do students who rely less on their phones report higher productivity?

Variables:

- Q5: Phone Addiction (Likert scale)
- Q4: Purposefulness (Likert scale)

Method:

- Spearman correlation to test if “addicted” users also rate screen time as meaningful

Visualization:

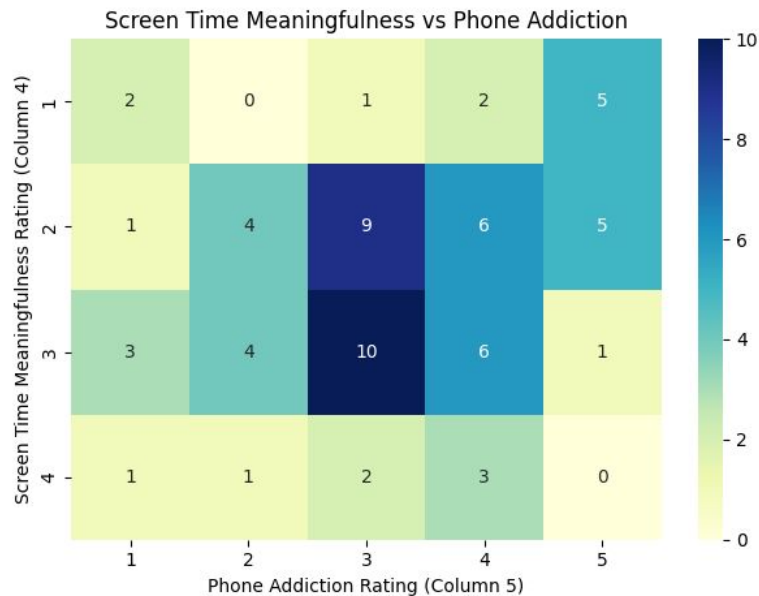
- Heatmap or scatterplot of addiction vs. meaningfulness ratings

Spearman Correlation

Spearman Correlation: -0.2613254608163534
P-value: 0.034053741694022845

- A Spearman correlation was performed to assess the relationship between phone reliance and GPA
- P-value $< 0.05 \rightarrow$ statistically significant

Interpretation: There is strong enough evidence to conclude a negative relationship between phone reliance and GPA. As students phone reliance increases, their GPA tends to decrease.



Hypothesis Testing

Hypothesis Test	Method	Visual	p-value threshold	Significance
Screen Time → GPA	Kruskal-Wallis	Boxplot	0.05	If $p < .05$, screen time group matters
Meaningfulness → GPA	Spearman Correlation	Scatterplot	0.05	If $p < .05$, more meaningful use links to higher GPA

Hypothesis Test 1: Screen Time vs. GPA

Research Question: Does the amount of daily screen time correlate with lower academic performance?

Variables:

- Independent variable: Q1 – Average daily screen time (ordinal categorical)
- Dependent variable: Q13 – Cumulative GPA (continuous numerical)

Hypotheses:

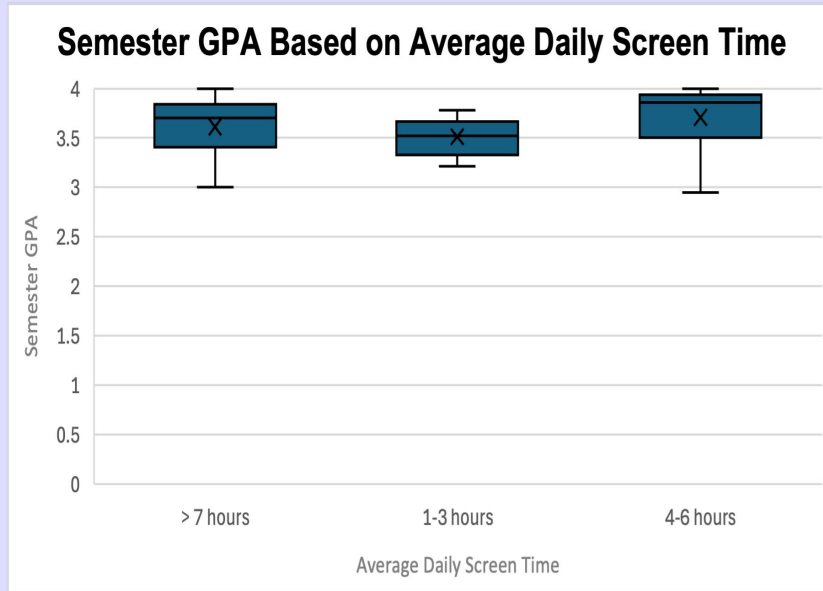
- Null (H_0): There is no significant difference in GPA across screen time groups.
- Alternative (H_1): GPA significantly differs across screen time groups.

Method: Kruskal-Wallis H test with significance level: $\alpha = 0.05$

Visualization: Boxplot: GPA grouped by Screen Time category

Results & Interpretation

If the p-value is < 0.05 , reject H_0 . That means screen time does significantly affect GPA. If $> 7\text{hr/day}$ group shows visibly lower median GPA, that supports our conclusion.



P-value: 0.204

Conclusion: Fail to reject null hypothesis, there is no statistically significant GPA difference between groups of average daily screen time.

Hypothesis Test 2: Meaningful Screen Time vs. GPA

Research Question: Does rating screen time as meaningful correlate with higher academic performance?

Variables:

- Independent variable: Q4 – Screen time meaningfulness (Likert: 1–5)
- Dependent variable: Q13 – GPA (numeric)

Hypotheses:

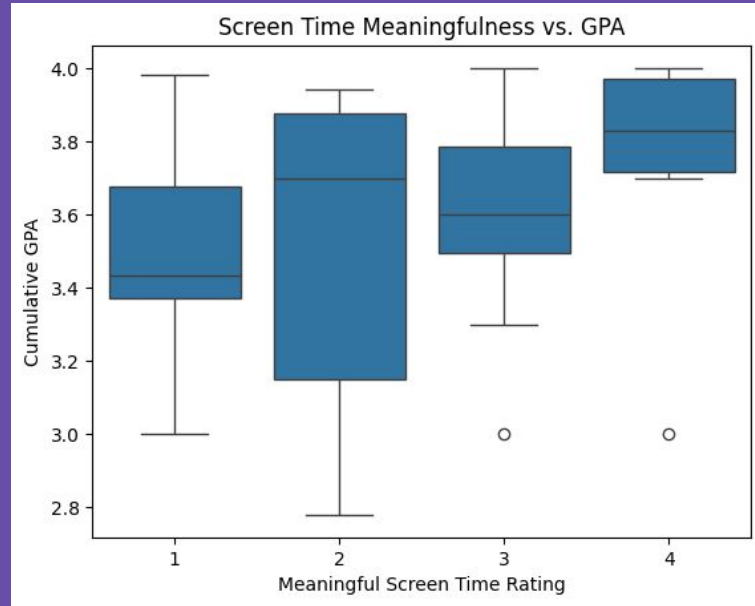
- Null (H_0): There is no correlation between screen time meaningfulness and GPA.
- Alternative (H_1): There is a positive correlation between meaningful screen time and GPA.

Method: Spearman's rank correlation with significance level: $\alpha = 0.05$

Visualization: Scatterplot or jitter plot with regression line: GPA vs. Screen Time Meaningfulness

Results & Interpretation

If Spearman's $\rho > 0$ and $p\text{-value} < 0.05$, then that means that more meaningful screen time is significantly associated with higher GPA.



Rho: 0.204

P-value: 0.103

Conclusion: Fail to reject the null hypothesis, there is no statistically significant correlation between GPA and meaningfulness of screen time.

Future Steps



Collect more diverse data to improve the sample



Allow for numerical data instead of ordinal or categorical data



Conduct more statistical analysis and hypothesis testing with other variables included in the question responses

Insights

Key Insight

- Most results were not statistically significant, showing that screen time is complex.
- It's not just how much time students spend on their phones, but how they use them.

Main Finding

- A significant negative link was found between phone reliance and GPA.
- Students who feel more dependent on their phones tend to have lower GPAs.

What Makes Our Study Different

- Unlike studies focused only on social media, we looked at overall phone use and habits.
- This shows that self-perception and phone habits matter more than just screen time.

Conclusion

The analysis conducted tells us that of the UMD students that were surveyed, they manage their screen time effectively and they did not let it affect their overall academic performance.

However, the way that they used their phones rather than their total screen time impacted their academics and productivity.

Addressing phone reliance is the key to improving academic performance.

It is important to note that due to the small sample size of 67 respondents among the 41,000+ students at this university, this conclusion cannot be used to generalize the whole population of UMD students.

THANK YOU